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Graphene Metasurfaces-Based Surface Plasmon Resonance Biosensor for Virus Detection with Sensitivity Enhancement Using Perovskite Materials

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Abstract

This paper introduces a biosensor based on graphene metasurfaces, designed for virus detection in the terahertz (THz) regime. The proposed sensor comprises four resonators arranged in a semicircular configuration, strategically engineered to achieve enhanced sensitivity and overall performance. Computational simulations using COMSOL Multiphysics version 6.2 were employed to optimize geometric parameters and analyze the sensor's behavior. By integrating Au, SrTiO₃, graphene, and black phosphorus, the biosensor exhibits remarkable sensitivity to refractive index (RI) variations associated with various viruses. The maximum sensitivity demonstrated by the sensor is 4556 GHzRIU⁻¹. Other remarkable performance metrics include a figure of merit of 8.499 RIU⁻¹, a quality factor of 1.131, and a minimum detection limit of 0.149. Electric field distribution analysis reveals optimal absorption at 0.4 THz. Furthermore, the biosensor demonstrates the potential for 2-bit encoding applications. Compared to existing designs, the proposed biosensor offers significantly higher sensitivity for virus detection. The integration of advanced nanomaterials and metasurface design principles presents a promising avenue for rapid, label-free virus sensing, with potential applications in healthcare and biosecurity.

Keywords Surface plasmon resonance · Graphene · Metamaterial · Perovskite materials · 2-bit encoding · Nanophotonics · Label-free detection

Introduction

In the biomedical field, the identification of viruses is very crucial and much more if it is done so quickly. Starting from virus identification and through the various steps to the containment and treatment measures, the success of the virus detection techniques determines the results in disease control [1]. Studies in biotechnology and nanotechnology in recent

years have brought about new techniques to identify viral diseases from the basic techniques to modern ones [2]. Antiviral detection techniques include a vast category of strategies ranging from conventional to the modern technological methodologies [3, 4] and culturing techniques, which although traditional, are still valid in clinical and research laboratories. New technologies which are built upon the development of nanotechnology, biosensors on a chip, and lab-on-a-chip devices allow for rapid and sensitive detection [5]. Furthermore, optical and spectroscopic techniques such as Raman spectroscopy and fluorescence-based methods are label-free, high-throughput approaches to viral recognition [6, 7].

Over the last few years, terahertz (THz) spectroscopy has been attracting interest in biosensing applications [8]. THz radiation has the ability to interact with biological substances and provides structural and functional information [9]. This makes them safer for biological tissues compared to X-rays and gamma rays [10]. This capability is especially beneficial for the applications in the areas of health care and biological security where fast and reliable identification is a must. On the other hand, PCR and ELISA techniques are

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followed by the process of sample preparation and labeling which is time-consuming and tedious [11].

Nanotechnology in the past few years has also produced materials that greatly expand the abilities in the use of THz sensing. Elements like graphene, perovskite materials, plasmonics materials, carbon nanotubes, quantum dots, and metamaterials have certain characteristics that can be controlled and used as highly effective sensors for virus detection [12, 13]. For instance, the high carrier mobility of graphene and tunable optical response allows for efficient THz wave control [14–16], whereas quantum dots provide excellence light-matter interaction at the terahertz range [17]. To promote the integration of nanomaterials to biosensor designs, understanding the material characteristics and properties is critical to improving the biosensor's effectiveness [18].

Biosensors have demonstrated great impacts in healthcare, environment, and food control [19, 20]. The implementation of biosensors combines biological recognition components with transducing techniques to quantify analytes accurately and at a high speed [21, 22]. The core of any biosensor is the integration of biological molecules, for instance, enzymes, antibodies, nucleic acids, or whole cells, and physical transducers that translate biological interactions into measurable signals [23]. This coupling enables the direct quantitation of the target analytes; this is because most methods do not require sample preparation steps, which make it possible to practice on-site [24]. Biosensors are universal in functionality, that is, it is possible to detect virtually any target.

Metasurfaces on the other hand are composed of meta-atoms and demonstrate optical properties that can be finely tuned to interact with specific biomolecules with high sensitivity [25]. The integration of black phosphorus, gold nanoparticles, and graphene within a metamaterial framework presents a synergistic approach of enhancing sensor performance [26, 27]. Black phosphorus is a two-dimensional material that offers tunable bandgap properties and high carrier mobility which facilitates efficient light absorption and generation of localized surface plasmons in the THz regime [28]. Gold nanoparticles further enhance sensitivity through their plasmonic resonances, enabling precise detection and quantification of biomolecular interactions [27, 29, 30].

Graphene metasurface SPR biosensors have diverse uses across multiple domains. In medicine, these highly sensitive devices enable quick detection of virus-causing agents. This capability is vital for early identification and prompt medical intervention, playing a key role in managing disease outbreaks and averting widespread health crises [19]. In environmental monitoring, the sensors can detect trace levels of pollutants and hazardous biological agents, ensuring public safety and environmental protection [31]. In the food industry, they can be used for detecting contamination and ensuring food safety by identifying pathogens and

spoilage organisms [32]. Moreover, in the field of biosecurity, these sensors can play a critical role in the detection of bioterrorism agents, providing a robust and rapid response mechanism to potential threats [33]. Key studies, such as those by Bhishma et al. [34] and Nasih et al. [35] among others have demonstrated the role of biosensor for a wide range of applications.

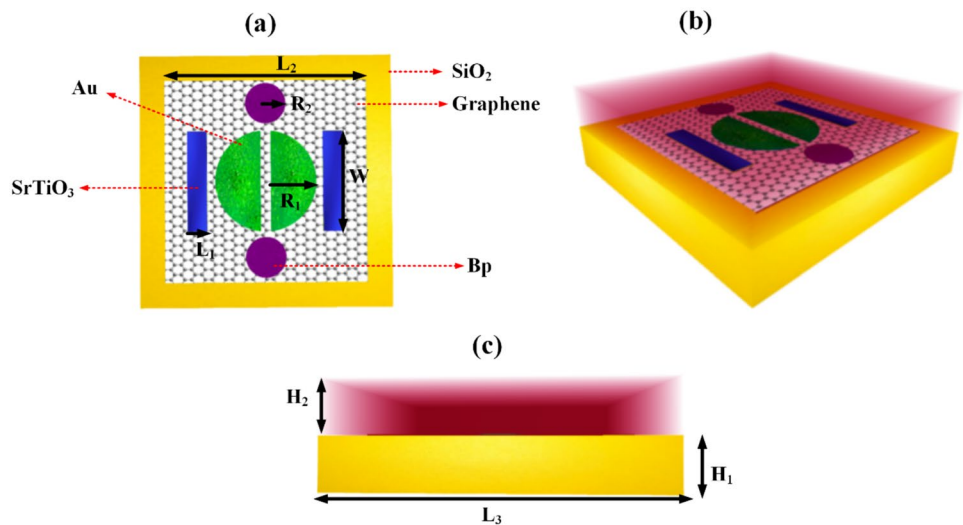
This paper explores the potential and effectiveness of a biosensor based on graphene metasurfaces for detecting viruses. By combining advanced materials and employing simulation-driven design, we aim to enhance sensitivity and other performance metrics. The use of SiO₂ as the substrate offers structural support, optical transparency, and compatibility with microfabrication techniques, which are crucial for incorporating other metamaterial designs. The sensor's performance is thoroughly analyzed and simulated using COMSOL Multiphysics.

Proposed Sensor Design

The proposed sensor design is based on a simple configuration comprising four resonators engineered to achieve high sensitivity and performance in the detection of different viruses. Central to the biosensor design is a resonator structured in the form of two semicircles, each with a radius of 2500 nm (R_1). In the other case, the semicircular resonators are affixed to a square-shaped resonator measuring 8500 nm (L_2). This arrangement enhances the sensor's ability to interact with target analytes effectively. Surrounding the semicircular resonators are other resonator elements, including rectangular and circular shapes. The rectangular resonator measures 2000 nm (L_1) by 500 nm (W). These resonators collectively create a simple and precisely controlled electromagnetic environment crucial for sensitive detection. To ensure robustness and operational reliability, the resulting resonator assembly is securely mounted on a SiO₂ substrate. This substrate not only provides mechanical support but also facilitates seamless integration into larger electronic arrays, thereby enabling scalable sensor fabrication for practical deployment. Figure 1a–c in the design documentation offer comprehensive visual representations from various viewing angles, illustrating the simple arrangement and spatial relationships between the resonators. It is important to note that while these figures provide detailed insights into the design, they are schematic and do not accurately depict the relative sizes of the components in real cases. Additionally, the sensor dimensions pertinent to its operational effectiveness include a height of the analyte at 1530 nm (H_2) and the overall dimensions of the biosensor at 10,080 nm (L_3) and 1520 nm (H_1).

For the purposes of constructing the metasurface-based sensor depicted in Fig. 1, a material that provides mechanical

Fig. 1 A conceptual design of a metasurface-based sensor. It provides multiple perspectives of the sensor. An overhead view (a). A three-dimensional representation (b). A front-facing view (c)



support and compatibility with electronic arrays is needed; this material is patterned to form silicon dioxide (SiO_2) substrate. The resonators formed as semicircles with a radius of 2500 nm, squares of 8500 nm, rectangles of 2000 nm by 500 nm, and other resonator geometries including circles are defined on this substrate by Electron Beam Lithography (EBL). After patterning, a metal like gold and Bp are sputtered or in other words evaporated and condensed on the corresponding resonators. The next process involves fine patterning by processes like the reactive ion etching or ion milling to carve the resonators. During the process, major focus is put on dimensions and alignment so as to maximize the interaction of the resonators with the target analytes to enhance the detection of the virus. After fabrication, the quality control includes optimizing microscopy to confirm the patterns and dimensions in relation to the sensor. The last layer of the sensor assembly is put to wrap up the sensor from influences of the environment in order to enable further real-life applicability.

$$n(\lambda) = \sqrt{1 + \frac{B_1 C}{\lambda^2 - C_1} + \frac{B_2 \lambda^2}{\lambda^2 - C_2} + \frac{B_3 \lambda^2}{\lambda^2 - C_3}} \quad (1)$$

$$\epsilon_{\text{Au}} = \epsilon_{\infty} - \frac{\omega_D^2}{\omega(\omega + j\gamma_D)} - \frac{\Delta_e \Omega_L^2}{(\omega^2 - \Omega_L^2) + j\Gamma_L \omega} \quad (2)$$

Equation 1 is used to calculate the refractive index of SiO_2 , while Eq. 2 is used to find the dielectric constant of gold.

Graphene Overview

One of the most interesting features about graphene is that it possesses a massless-like electron behavior. In most

materials, electrons are like positively charged objects within the atomic framework or lattice; they have mass when passing through the lattice [36]. Their movement is characterized by equations that take into consideration this mass, that defines the manner in which these entities move within the material [37]. However, graphene stands out uniquely because it forms various distinct moiré patterns through its interaction with the underlying layer [38]. Another feature is its Dirac cone [39–43]. This is a very special way of drawing the energy momentum map in which the conduction and the valence bands touch at a single point, the Dirac point. In these materials, the dependence of energy on momentum is linear, in contrast to the values of the majority of materials, at a Dirac point near which this dependence is a straight line [44]. As a result of such, linear dispersion, electrons in graphene move as if they are true relativistic particles so like photons with no relativistic mass at rest. They are very peculiar and receive the name of “Dirac fermions.” Such electrons do not possess a true mass and can travel at extremely high velocities of up to 1/300 the velocity of light, which compares well to traditional ferromagnetic semiconductors. The massless-like behavior also implies that electrons are not as affected by the lattice as they undergo little scattering and less energy dissipation [45]. Consequently, it can be predicted that the electron conductance in graphene would depend on a number of factors and can be very efficient in enabling an electric current to flow, based on the following reasons. This massless-like behavior is a quantum mechanical phenomenon due to the characteristics of the graphene sheet [46–49]. It is one of the main reasons that graphene has such impressive electrical characteristics and is such a material of focus for possible electronic uses [50]. Equations 3–6 detail the conductivity of graphene [51–54].

$$\varepsilon(\omega) = 1 + \frac{\sigma_s}{\varepsilon_0 \omega \nabla} \quad (3) \quad \text{SNR} = \frac{\Delta f}{\text{FWHM}} \quad (12)$$

$$\sigma_{\text{intra}} = \frac{-je^2 k_B T}{\pi \hbar^2 (\omega - j2r)} \left(\frac{\mu_c}{k_B T} + 2 \ln \left(e^{\frac{\mu_c}{k_B T}} + 1 \right) \right) \quad (4) \quad \text{SR} = S \times \text{DL} \quad (13)$$

$$\sigma_{\text{inter}} = \frac{-je^2}{4\pi \hbar} \ln \left(\frac{2|\mu_c| - (\omega - j2r)\hbar}{2|\mu_c| + (\omega - j2r)\hbar} \right) \quad (5) \quad \text{DA} = \frac{1}{\text{FWHM}} \quad (14)$$

$$\sigma_s = \sigma_{\text{intra}} + \sigma_{\text{inter}} \quad (6)$$

Graphene has been incorporated in this research to boost the efficiency of the surface plasmon resonance (SPR) biosensor that we have configured. Graphene is chosen because it comprises unprecedented electro-, opto-, and mechanical performances that form the basis for biosensing [55, 56]. The 2D structure of graphene is one of the most advantageous features because the density of the active sites where biomolecules can be attached to is relatively high due to a high surface-to-volume ratio [57]. The array of these active sites has to be high because it facilitates better adsorption of the target analytes by increasing contact between the sensor surface and the analytes which is significant in increasing the sensitivity of the sensor [53, 54, 58]. Furthermore, graphene has high electrical conductivity, and plasmonic effect which further improves the SPR effect [59]. Graphene excels in high electron mobility, when it comes to enhancing plasmonic resonance in combination with metallic particles—in our case gold. A strong plasmonic resonance is observed due to which the local electromagnetic field near the sensor surface is increased, which when analyte molecules interact with the sensor surface, results in a more significant change in the refractive index. This amplification effect is very important in enhancing the sensitivity of the biosensor [60–62].

The effectiveness of every sensor depends on different measures. Some of these measures comprise [63]:

$$S = \frac{\Delta f}{\Delta n} \quad (7)$$

$$\text{FOM} = \frac{S}{\text{FWHM}} \quad (8)$$

$$Q = \frac{fr}{\text{FWHM}} \quad (9)$$

$$\text{DL} = \left(\frac{\Delta n}{1.5} \right) \times \left(\frac{\text{FWHM}}{\Delta f} \right)^{1.25} \quad (10)$$

$$\text{DR} = \frac{f_r}{\sqrt{\text{FWHM}}} \quad (11)$$

Virus Sensing Based on Refractive Indices

Technique for virus sensing based on refractive index changes has been reported as one of the most effective in diagnostic technique in the management of viral diseases. Since viruses are nanoscale particles, optical properties of viruses are affected by their size and components. This changes the refractive index of the media surrounding the particles and forms the basis for their identification. The virus sensing through the refractive indices relies on the optical techniques like the SPR or photonic crystals. These methods are based on the use of such a peculiarity of viruses as the ability to change the refractive index of the surface of the sensor or the surrounding environment. This can in turn be measured as a change of resonance frequency or intensity referring to concentration or sort of virus. Some of the benefits that can be obtained by using the refractive index changes to detect viruses include the determination of slight changes in refractive index due to virus binding or interaction; one can obtain high sensitivity and specificity within short time intervals and without the need for further sample processing and additional chemicals. Uses of this technology are across the entire healthcare sectors including diagnosis of viral diseases where quick identification of the virus is vital in the treatment and prevention of further spread, and in environmental scanning and security. Despite the current advancements, future research continues to focus on improving the performance metrics such as sensitivity and specificity of the sensing platforms and, therefore, are tremendously valuable in clinical and research applications.

Graphene, gold, and perovskite materials are chosen for this research because each of them holds a significant role in the improvement of the biosensor's sensitivity. Graphene is preferred due to its outstanding electronic features such as high electron mobility, large surface area, and mechanical strength in improving the biosensors' sensitivity. It also has a two-dimensional design that enables the capturing of a large number of biomolecules to interact with target species. In the same regard, the conductivity as well as plasmonic features of graphene are advantageous in SPR sensors, resulting to improved sensitivity of the material in the detection of refractive index variations due to the presence of viruses. On the other hand, gold is preferred in the

development of SPR sensors because of its great plasmonic characteristic. Gold nanoparticles amplify the local electromagnetic field when exposed to light; thus, the SPR signal is boosted. This amplification effect is very useful when there is a need to determine small changes in the refractive index required for virus detection. Furthermore, the gold electrode serves a dual purpose by offering a stable surface to get biomolecules immobilized on the electrode matrix and ensuring the sensor's efficiency in biological systems does not decline due to reactions with the environment. Additional components include perovskite materials that are added to increase the sensitivity of the created sensor, strontium titanate (SrTiO₃) for instance. These materials have specially high dielectric constants and non-linear optical properties combined with strong light-matter interaction. Perovskites enhance the architectonics of sensor by boosting the efficiency of light trapping in graphene and gold besides enhancing the plasmonic resonance. This in turn leads to an enhancement of the sensor's performance parameters for the detection of viruses.

Results and Discussion

We have employed COMSOL Multiphysics 6.2 to design the proposed sensor structure. The design which incorporates a metasurface is taken through extensive analysis involving adjustments to its geometric parameters to optimize sensitivity among other performance parameters. This evaluation narrowed on evaluating transmittance as well as studying the electric field distribution across the sensor.

Variation of Graphene Chemical Potential (GCP)

At first, we assessed how the GCP impacts the transmittance response of the sensor design. Figure 2a and b depict the influence of the GCP on the transmittance characteristics of the sensor design. The GCP denotes the energy difference

between graphene's Fermi level and its Dirac point, a critical parameter affecting its electronic behavior. Figure 2a displays a line graph illustrating the relationship between transmittance and various GCP values, showing a consistent decline in transmittance as GCP increases. Figure 2b presents a color-coded spectral representation indicating a rightward shift in transmittance response with increasing GCP. Notably, negligible reductions in transmittance are observed at lower GCP values such as 0.1 eV, 0.2 eV, and 0.3 eV. In contrast, significant decreases in transmittance are observed across other GCP values within the 0.1–0.4 THz frequency range. In this analysis, transmittance reductions of 98.39%, 98.23%, 97.58%, 90.57%, 86.49%, 81.95%, 77.17%, 72.33%, and 67.53% are documented within this frequency band. Changes in the GCP lead to significant decreases in transmittance because of alterations in the electronic properties of graphene. The GCP, representing the energy difference between graphene's Fermi level and its Dirac point, plays a crucial role in its electrical conductivity. As the GCP levels increase, the Fermi level shifts away from the Dirac point, thereby modifying the band structure around this point. This alteration affects how electronic states involved in the transmission of light or electromagnetic waves through graphene are characterized. The specific relationship between GCP and carrier density is such that the materials with higher GCP have a higher carrier density and therefore these materials have a higher ability to absorb and scatter the incident light or electromagnetic waves in the material.

Optimization of Parameters

In this section, further structural optimization is undertaken to refine the design for optimal performance. This comprehensive process involves adjusting various parameters, including fine-tuning the angle of incidence and various resonators' dimensions, and carefully determining the ideal parameters. These refinements are crucial for enhancing the efficiency and effectiveness of the sensor design.

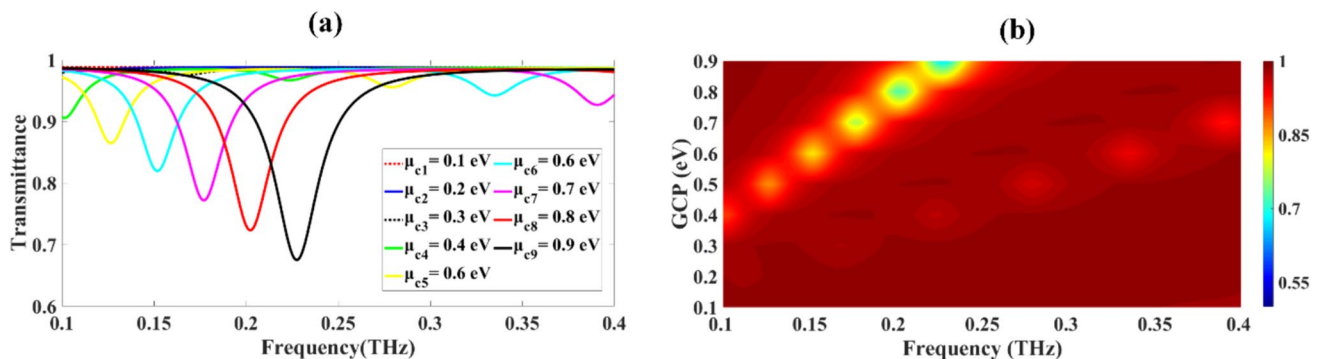


Fig. 2 **a** Alterations in transmittance response consequent to variations in GCP. **b** A Fermi plot delineating the same results

The preliminary study examined the impact of varying the size of two circular resonators on the transmission characteristics of the proposed sensor. The results of this analysis are shown in Fig. 3a and b. Figure 3a illustrates the concurrent adjustment of both resonators' sizes, ranging from 300 to 700 nm in steps of 100 nm. The observed drops in transmittance were 64.63%, 65.42%, 66.98%, 71.07%, and 80.29%. The Fermi plot in Fig. 3b demonstrates that larger resonator sizes lead to a shift of the transmittance response towards lower frequencies. The subsequent investigation focused on varying the size of the square resonator and examining how it affects the transmittance characteristics of the proposed sensor. The results are shown in Fig. 3c and d. Figure 3c illustrates how changes in the resonator dimensions, ranging from 7000 to 9000 nm in 500 nm increments, impact transmittance. The observed decreases in transmittance were 87.76%, 84.32%, 81.67%, 79.32%, and 77.61%. Furthermore, Fig. 3d clearly demonstrates that increasing the resonator size shifts the transmittance response towards higher frequencies on the Fermi plot.

A systematic investigation was carried out to examine how the dimensions of semi-circular resonators affect optical transmittance. The study involved varying the dimensions from 1000 to 5000 nm in increments of 500 nm. The results of this analysis are presented in Fig. 4a and b. Figure 4a displays transmittance spectra over a frequency range of 0.4–0.6 THz, illustrating reductions in

transmittance of 90.49%, 90.20%, 90.36%, 90.18%, 90.33%, 90.57%, 87.76%, 87.76%, and 87.76% across different conditions. In contrast, Fig. 4b utilizes a color plot to show how transmittance spectra vary with changes in the dimensions of the two semi-circular resonators. This visual representation indicates a consistent overall transmittance magnitude, accompanied by a subtle rightward shift in spectral response as frequency increases. In our fourth study, we aimed to examine how the distance between two semi-circular resonators affects the reduction in transmittance. Figure 4c and d illustrate the results of our simulations, showing the corresponding behaviors. Throughout this investigation, we methodically adjusted the distance between 500 and 3000 nm in increments of 500 nm. The transmittance response was observed across the frequency band spanning 0.4 to 1 THz as depicted by the line plot in Fig. 4c. Notably, the transmittance drops achieved are as follows: 87.76%, 81.84%, 75.83%, 68.88%, 62.33%, and 55.57%. Additionally, the Fermi plot in Fig. 4d reveals that as the separation distance increases, the transmittance response shifts toward higher frequencies.

We further conducted a comprehensive analysis to determine the impact of rectangular resonator size changes on transmittance response. The investigation involved varying the resonator length between 500 and 1500 nm. Results were observed in the 0.1–0.3 THz frequency range, as illustrated in Fig. 5a and b. In a parallel

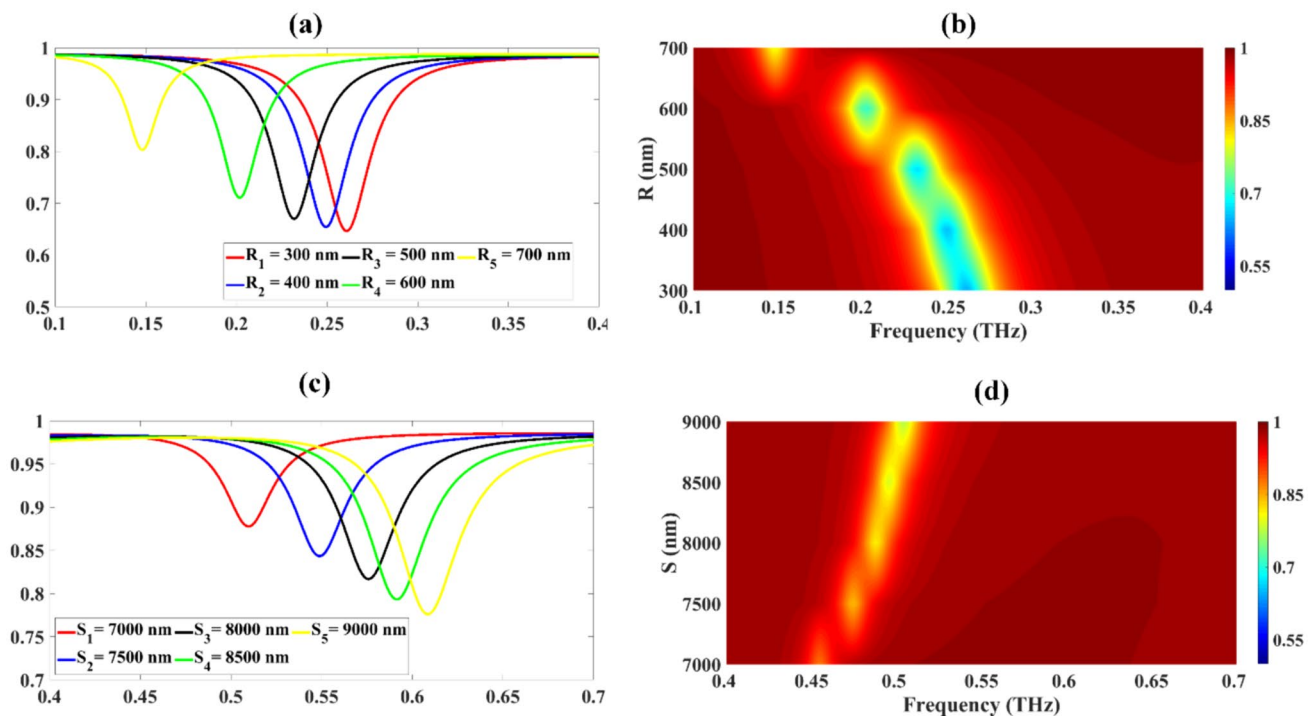


Fig. 3 It illustrates how changing the sizes of the two circular resonators impacts the transmittance responses depicted in **a** and **b**. It also shows how varying the dimensions of the square resonator affects the transmittance responses shown in **c** and **d**

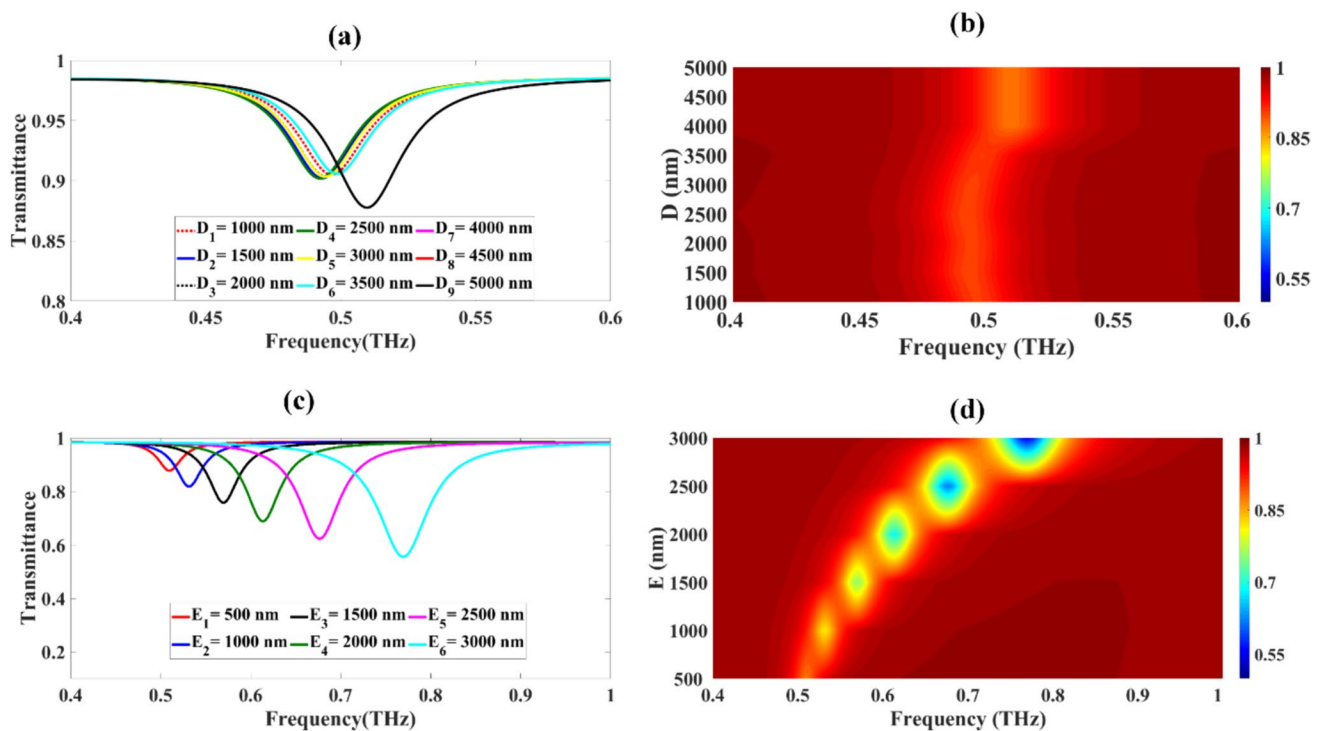


Fig. 4 A study examining how changes in the dimensions of semi-circular resonators as well as their distance of separation affect their transmittance characteristics

study, we adjusted the width from 2000 to 5000 nm. The complete results of this parametric study are displayed in Fig. 5c and d. Figure 5a and c illustrate the transmittance response to variations in length and width. Transmittance reductions of 67.46%, 67.85%, 68.56%, 69.97%, 72.00%, and 75.10% were observed for varying lengths, while width variations resulted in reductions of 58.91%, 59.01%, 59.26%, 59.85%, 61.26%, 63.68%, and 67.53%. Fermi plots depicted in Fig. 5b and d clearly indicate a leftward shift in transmittance response with increasing length and width. Our final analysis examined how changes in the incident angle influenced the transmittance response. The outcomes of this investigation are presented in Fig. 5e and f. The angle was varied from 0 to 80° at intervals of 10°. The angle variation is observable across the simulated range of 0.1–0.4 THz. The attained transmittance drops are 67.46%, 67.13%, 66.11%, 64.29%, 61.49%, 57.33%, 51.18%, 41.85%, and 26.81%.

Detection Analysis

This section examines the transmittance response for various virus types, focusing on the effects of refractive index (RI) variations and their correlation with concentration. Figure 6a–d illustrate the influence of RI alterations on the transmittance response for different viruses, resulting in distinct spectral dips. In Fig. 6a, transmittance minima of

5.42%, 4.97%, 4.62%, 4.34%, 4.10%, and 3.90% are observed at frequencies of 0.595, 0.427, 0.400, 0.377, 0.356, and 0.337 THz, respectively. Figure 6b demonstrates transmittance minima of 4.67%, 4.66%, 4.65%, 4.64%, 4.05%, and 3.85% at frequencies of 0.409, 0.403, 0.402, 0.361, 0.351, and 0.333 THz, respectively. In Fig. 6c, transmittance minima of 4.66%, 4.65%, 4.62%, 4.60%, 4.50%, 4.47%, and 4.43% are observed at frequencies of 0.403, 0.402, 0.401, 0.399, 0.390, 0.389, and 0.370 THz, respectively. Lastly, Fig. 6d exhibits transmittance minima of 44.70%, 4.68%, 4.67%, 4.65%, 4.64%, 4.63%, and 4.62% at frequencies of 0.420, 0.405, 0.404, 0.403, 0.402, 0.401, and 0.400 THz, respectively. These variations in transmittance spectra across different virus types demonstrate the sensitivity of THz spectroscopy to subtle changes in the refractive index of viral samples. The specific patterns of transmittance dips serve as spectral fingerprints, enabling the identification and differentiation of various virus types. Moreover, the correlation between these spectral features and virus concentration provides a basis for quantitative analysis. On the other hand, Fig. 7a–d depict the application of curve-fitting analysis aimed at exemplifying the relationship between resonance frequency and refractive indices (RIs). In this context, a linear regression model is employed initially to establish the correlation between resonance frequency and RI, exemplifying a corresponding decrease in transmittance as RI varies. The resultant linear regressions yield coefficients of determination (R^2) of 91.32%, 90.02%, 96.07%, and 90% for

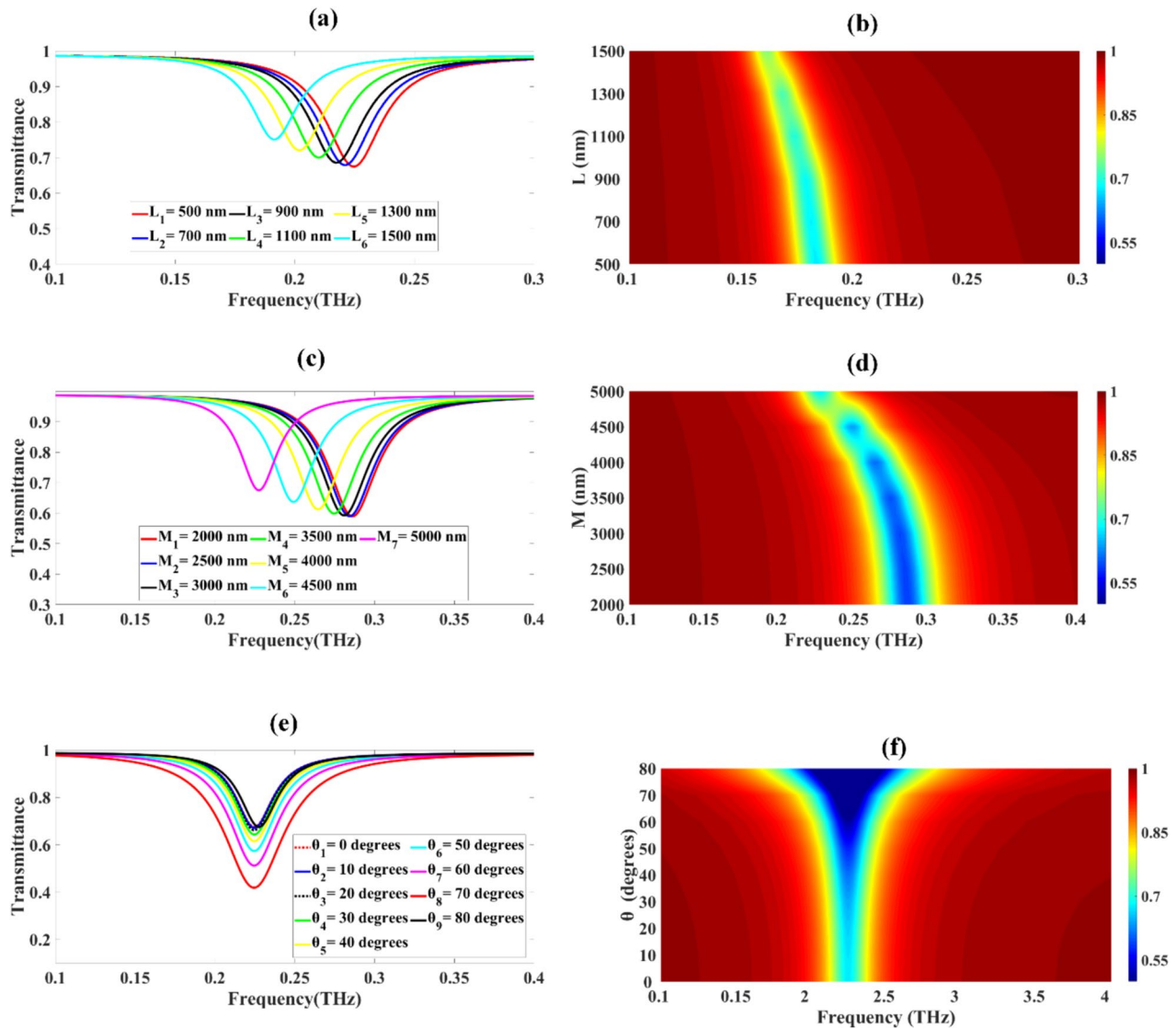


Fig. 5 An investigation into how altering the dimensions of rectangular resonators (length variation shown in **a** and **b**, width variation in **c** and **d**) and varying the angle of incidence (shown in **e** and **f**) influence the transmittance response of the proposed sensor design

Fig. 7a, b, c, and d, respectively, indicating a robust fit of the model to the data. The equations representing the linear fits are expressed as follows:

$$F = -0.2180RI + 0.7423 \quad (15)$$

$$F = -0.0921RI + 0.5209 \quad (16)$$

$$F = -0.1747RI + 0.6449 \quad (17)$$

$$F = -0.3175RI + 0.8419 \quad (18)$$

These equations describe the linear relationships derived from the plots presented in Fig. 7a–d, where F denotes resonance frequency and RI represents refractive index. The coefficients (-0.2180 , -0.0921 , -0.1747 ,

and -0.3175) indicate the slopes of the linear fits, illustrating the extent of change in resonance frequency per unit change in RI .

Electric Field Analysis

The electric field distributions resulting from the proposed metasurface design are depicted in Figs. 8a–l and 9a–l for frequencies spanning 0.1 THz, 0.4 THz, and 1 THz. At 0.1 THz demonstrated in Fig. 8a and b and g and h as well as Fig. 9a and b and g and h, the electric field exhibits a notably uniform distribution with minimal concentration or scattering observed across the metasurface structure. The uniformity indicates a scenario where the absorption of electromagnetic energy is negligible, implying a high degree of

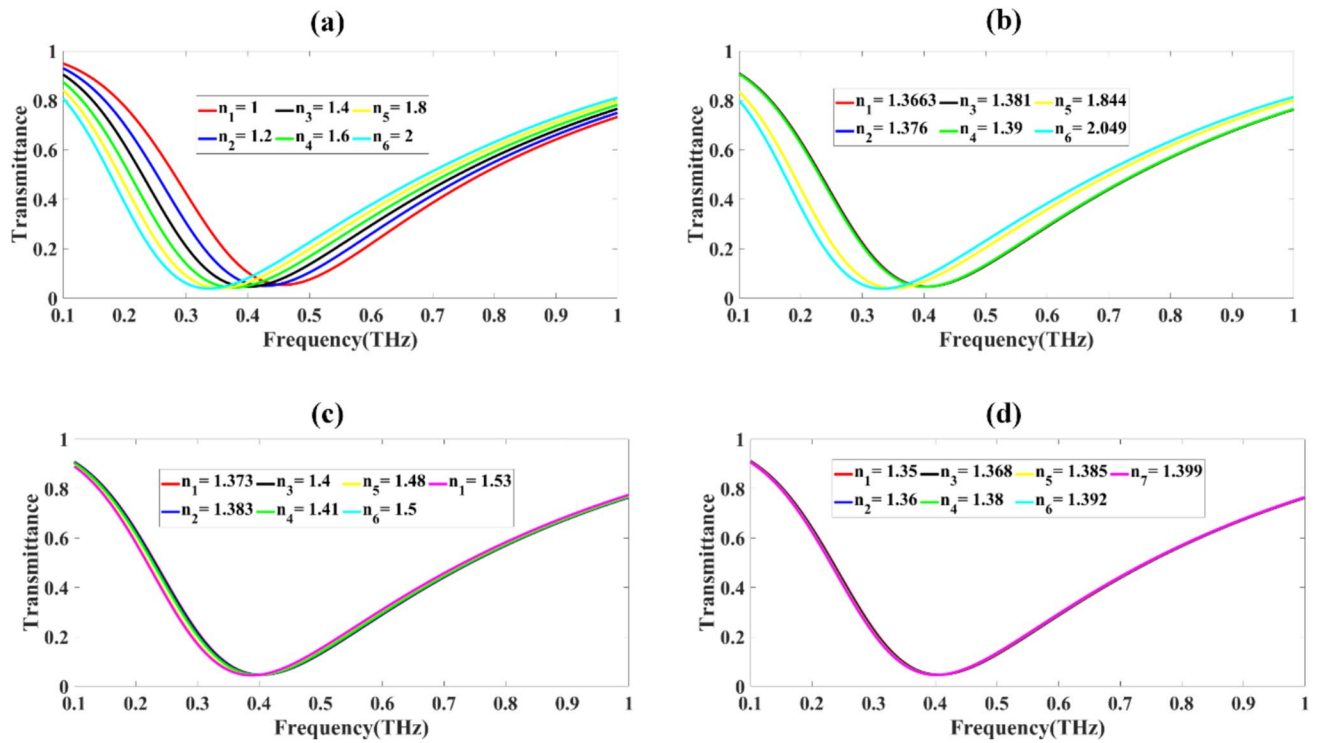


Fig. 6 a–d The transmittance outcomes of the proposed sensor for different refractive indices associated with various viruses, as specified in Table 1

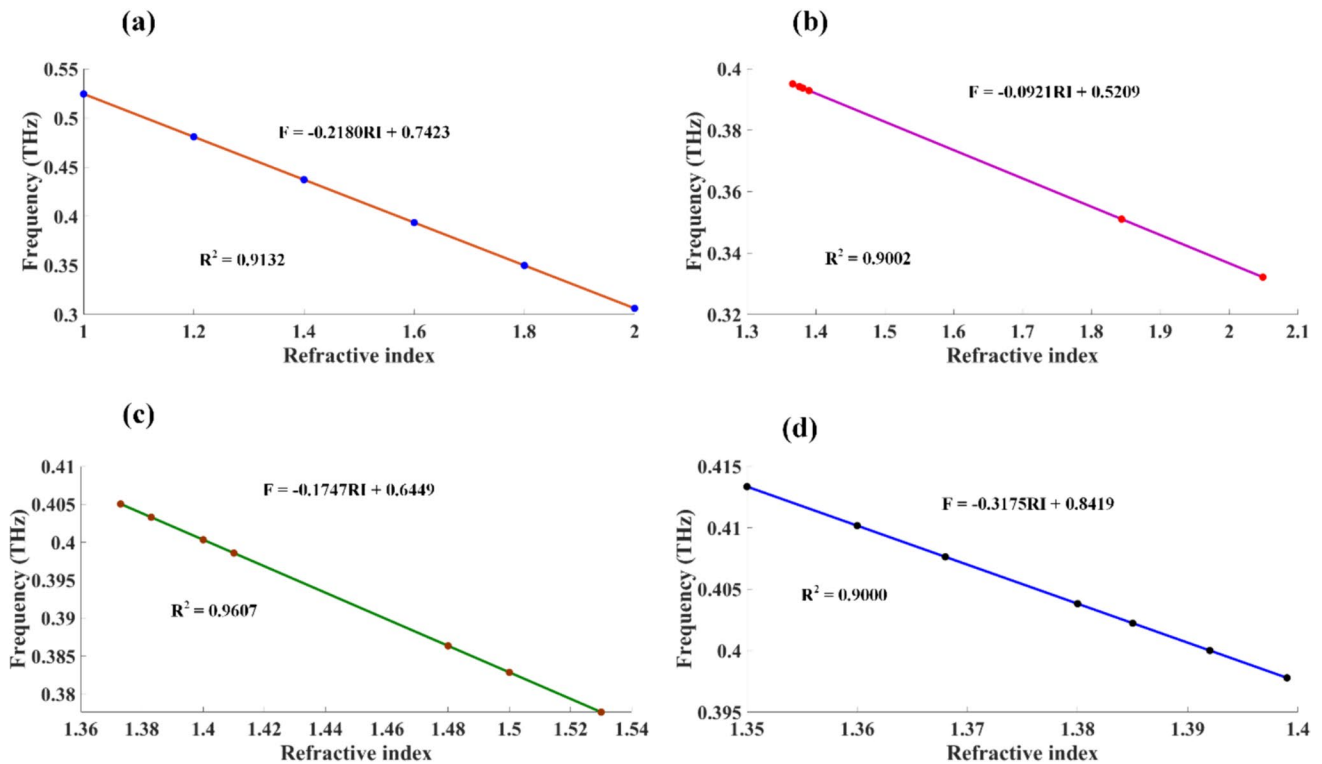
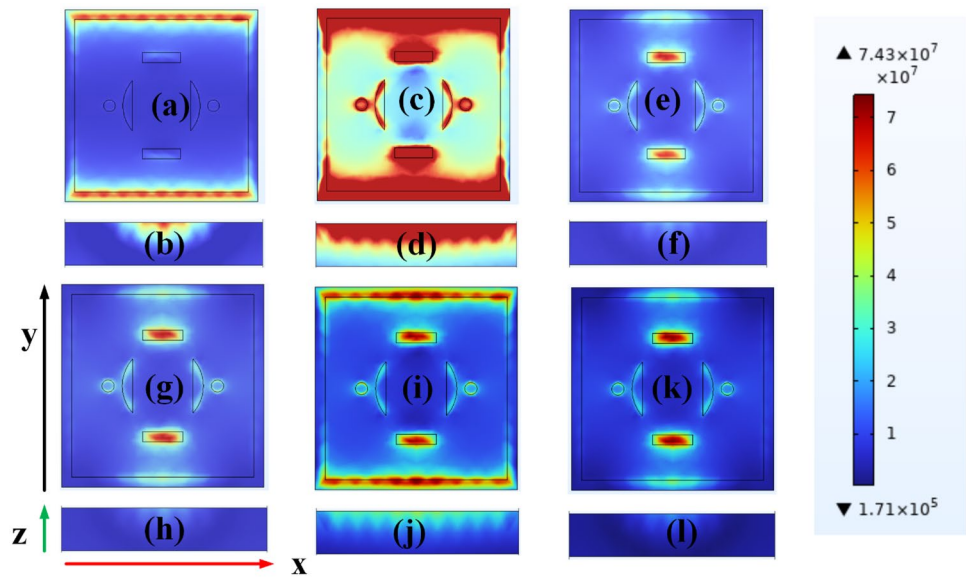


Fig. 7 a–d Four analyses of curve fitting. Each analysis involves fitting a linear model to the resonance frequency relative to the refractive index at which the transmittance drop is attained

Fig. 8 The results of the electric field from the graphene sensor in two different cases, corresponding to Fig. 6a and b. Each scenario is analyzed at three specific frequencies: **a, b** 0.1 THz; **c, d** 0.4 THz; and **e, f** 1 THz for the first scenario in Fig. 6a and **g, h** 0.1 THz; **i, j** 0.4 THz; and **k, l** 1 THz for the second scenario in Fig. 6b



transmittance through the material. As the frequency escalates to 0.4 THz exemplified in Fig. 8c and d and i and j, an observable increase in absorption becomes evident, as illustrated by a more pronounced scattering pattern of the electric field across the metasurface structure. This phenomenon suggests that the metasurface design effectively interacts with incident electromagnetic waves, resulting in enhanced absorption characteristics. The observed scattering pattern at this frequency indicates a heightened interaction between the electric field and the metasurface elements, facilitating greater energy absorption. Conversely, at 1 THz depicted in Fig. 8e and f and k and l as well as Fig. 9e and f and k and l, the electric field distribution exhibits characteristics akin to those observed at 0.1 THz, with minimal scattering

and absorption observed on the metasurface design. This similarity suggests consistent behavior of the metasurface across these frequencies, where electromagnetic energy experiences minimal impedance and significant transmittance through the material.

Tables 2, 3, 4, and 5 present results for the four virus detection scenarios listed in Table 1. In the initial case demonstrated in Table 2, the sensor achieves its peak sensitivity at 840 GHzRIU^{-1} . The second case records a maximum sensitivity of 4556 GHzRIU^{-1} . The third case demonstrates an optimum value of 633 GHzRIU^{-1} and the fourth case exemplifies maximum sensitivity of 1500 GHzRIU^{-1} . The sensor shows great sensitivity, particularly at higher frequencies, indicating its ability to detect even minor changes in

Fig. 9 The electric field findings from the graphene sensor under two specific scenarios are illustrated in Fig. 6c–d. We analyze three frequencies for each scenario: **a, b** 0.1 THz; **c, d** 0.4 THz; and **e, f** 1 THz for the first scenario in Fig. 6c and **g, h** 0.1 THz; **i, j** 0.4 THz; and **k, l** 1 THz for the second scenario in Fig. 6d

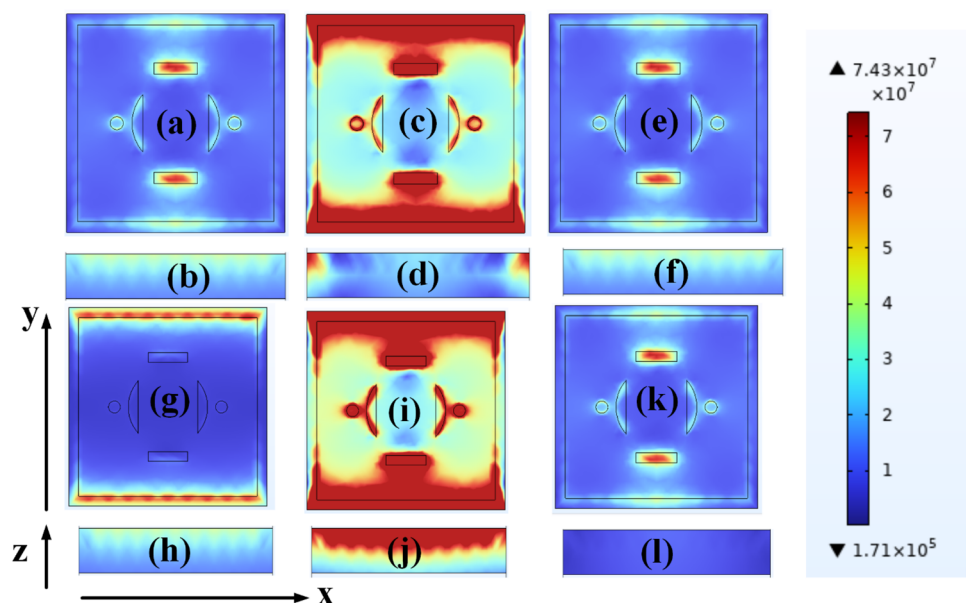


Table 1 Refractive indices of different cancerous and normal cells [64]

Virus	RI (<i>n</i>)	Virus	RI (<i>n</i>)	Virus	RI (<i>n</i>)
Cervical cancer cell	1.392	Human blood (healthy)	1.35	Dengue	1.4
Normal cervical cell	1.368	Breast cell (normal)	1.385	Malaria (n_2)	1.383
Jurkat (normal)	1.376	Basal cell (cancerous)	1.38	Malaria (n_1)	1.373
Jurkat (cancerous)	1.39	Basal cell (normal)	1.36	HIV	1.5
Normal breast cell	1.3663	Breast cell (cancerous)	1.399	-	-
PC12 cell	1.381	corona	1.53	-	-
HSC	1.844	Influenza	1.48	-	-
CSC	2.049	HSV	1.41	-	-

Table 2 The performance parameters of the proposed sensor design

<i>n</i> (RIU)	1	1.2	1.4	1.6	1.8	2
<i>f</i> (THz)	0.595	0.427	0.4	0.377	0.356	0.337
<i>df</i> (THz)		0.168	0.027	0.023	0.021	0.019
<i>dn</i> (RIU)		0.2	0.2	0.2	0.2	0.2
<i>S</i> (GHz/RIU)		840	135	115	105	95
FWHM (THz)	0.526	0.533	0.567	0.536	0.535	0.532
FOM (RIU ⁻¹)		1.576	0.238	0.215	0.196	0.179
<i>Q</i>	1.131	0.801	0.705	0.703	0.665	0.633
DL		0.565	5.994	6.827	7.631	8.588
DR	0.820	0.585	0.531	0.515	0.487	0.462
SNR		0.315	0.048	0.043	0.039	0.036
SR		0.474	0.809	0.785	0.801	0.816
DA	1.901	1.876	1.764	1.866	1.869	1.880
<i>X</i>		0.050	0.013	0.011	0.010	0.010

Table 3 The performance parameters of the proposed sensor design

<i>f</i> (THz)	0.409	0.403	0.402	0.361	0.351	0.333
<i>n</i> (RIU)	1.3663	1.376	1.381	1.39	1.844	2.049
<i>df</i> (THz)		0.006	0.001	0.041	0.01	0.018
<i>dn</i> (RIU)		0.0097	0.005	0.009	0.454	0.205
<i>S</i> (GHz/RIU)		619	200	4556	22	88
FWHM (THz)	0.536	0.536	0.536	0.536	0.535	0.535
FOM (RIU ⁻¹)		1.154	0.373	8.499	0.041	0.164
<i>Q</i>	0.763	0.752	0.750	0.674	0.656	0.622
DL		1.776	8.597	0.149	43.793	9.484
DR	0.559	0.550	0.549	0.493	0.480	0.455
SNR		0.011	0.002	0.076	0.019	0.034
SR		1.099	1.719	0.679	0.965	0.833
DA	1.866	1.866	1.866	1.866	1.869	1.869
<i>X</i>		0.004	0.001	0.017	0.006	0.009

Table 4 The performance parameters of the proposed sensor design

f (THz)	0.403	0.402	0.401	0.399	0.39	0.389	0.37
n (RIU)	1.373	1.383	1.4	1.41	1.48	1.5	1.53
dn (RIU)		0.01	0.017	0.01	0.07	0.02	0.03
Df (THz)		0.001	0.001	0.002	0.009	0.001	0.019
S (GHz/RIU)		100	59	200	129	50	633
FWHM (THz)	0.536	0.536	0.536	0.536	0.538	0.538	0.538
FOM (RIU ⁻¹)		0.187	0.110	0.373	0.239	0.093	1.177
Q	0.752	0.750	0.748	0.744	0.725	0.723	0.688
DL		17.194	29.229	7.229	7.757	34.547	1.306
DR		0.549	0.548	0.545	0.532	0.530	0.504
SNR		0.002	0.002	0.004	0.017	0.002	0.035
SR		1.719	1.719	1.446	0.997	1.727	0.827
DA	1.866	1.866	1.866	1.866	1.859	1.859	1.859
X		0.001	0.001	0.002	0.006	0.001	0.010

the refractive index. The resonance peaks are sharp, as evidenced by the full width at half maximum (FWHM) values of 526 GHz, 535 GHz, 536 GHz, and 535 GHz for the cases presented in Tables 2, 3, and 4, respectively. A narrower FWHM suggests more precise detection of spectral features related to biomarkers of viruses. The figure of merit (FOM), calculated as the sensitivity-to-FWHM ratio, ranges from 0.179 to 0.156 RIU⁻¹ for cases 1 to 4, indicating varying degrees of sensor performance across these scenarios. The quality factor (Q) is another important parameter. Higher Q values are indicative of sharper, more selective resonance peaks that enhance the sensor's ability to discern small spectral shifts. For the four cases, the maximum quality factors attained are 1.131, 0.763, 0.752, and 0.785. The sensor's detection limit (DL) quantifies its capability to detect minimal changes in refractive index. Lower DL values signify higher sensitivity, suggesting that this sensor can reliably detect even slight variations in RIs. The minimum detection limits for the four cases are 0.565, 0.149, 1.306, and 0.581.

The additional performance metrics of the sensor under consideration are shown in Tables 2, 3, 4, and 5. Additionally, we performed a comparative analysis between our design and other existing cases documented in the literature as exemplified in Table 6.

The comparison shows that the proposed sensor demonstrates good results compared to findings in other studies. Particularly noteworthy is its sensitivity of 4556 GHz/RIU. Although certain designs in literature excel in some performance parameters, the sensor proposed here, which integrates Au, SrTiO₃, graphene, and black phosphorus, presents distinct advantages for virus sensing applications.

The simulation results demonstrating how well the proposed sensor can encode information are shown in Fig. 10a–d. Different values of μ_{c1} and μ_{c2} lead to different levels of transmittance. When both μ_{c1} and μ_{c2} are set to 0.1 eV, Fig. 10d displays a clear reduction in transmittance. Conversely, in Fig. 10c, where both parameters are increased to 0.9 eV, close to perfect transmittance is

Table 5 The performance parameters of the proposed sensor design

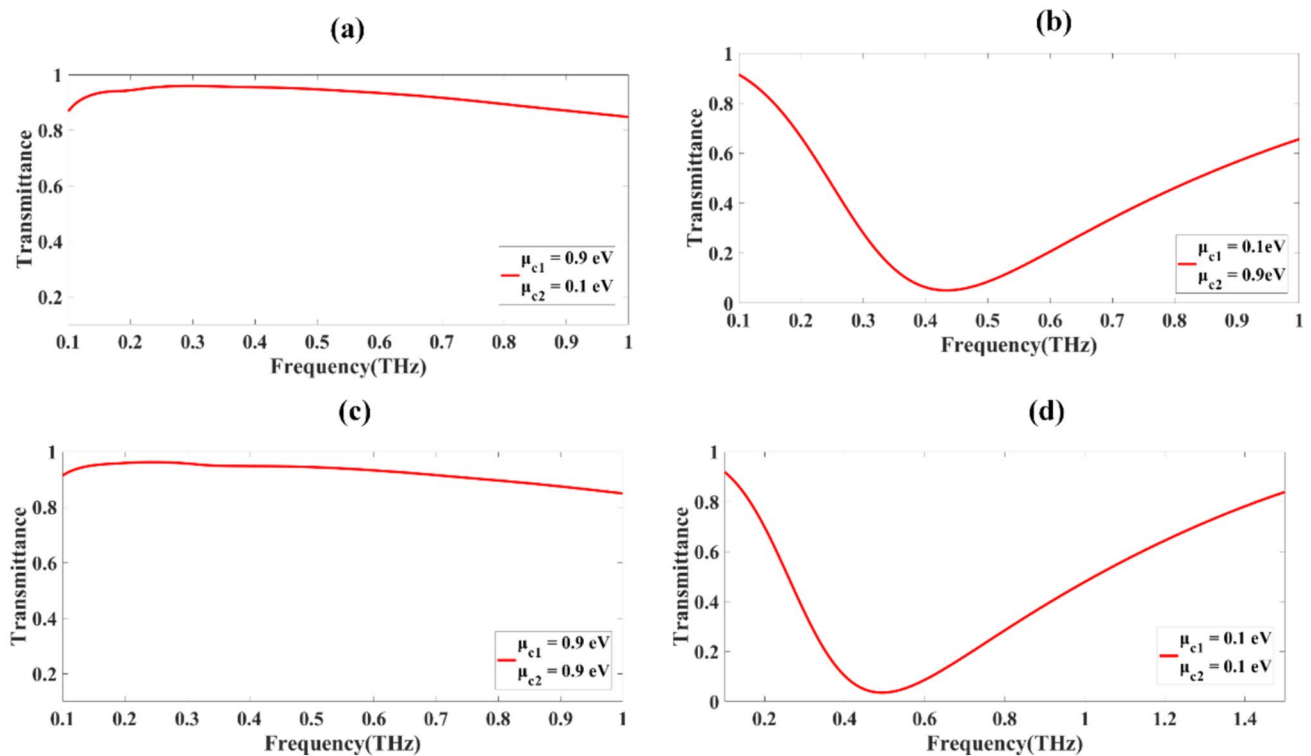
f (THz)	0.42	0.405	0.404	0.403	0.402	0.401	0.4
n (RIU)	1.35	1.36	1.368	1.38	1.385	1.392	1.399
df (THz)		0.015	0.001	0.001	0.001	0.001	0.001
dn (RIU)		0.01	0.008	0.012	0.005	0.007	0.007
S (GHz/RIU)		1500	125	83	200	143	143
FWHM (THz)	0.535	0.535	0.535	0.535	0.535	0.535	0.535
FOM (RIU ⁻¹)		2.804	0.234	0.156	0.374	0.267	0.267
Q	0.785	0.757	0.755	0.753	0.751	0.750	0.748
DL		0.581	13.723	20.584	8.577	12.007	12.007
DR	0.574	0.554	0.552	0.551	0.550	0.548	0.547
SNR		0.028	0.002	0.002	0.002	0.002	0.002
SR		0.872	1.715	1.715	1.715	1.715	1.715
DA	1.869	1.869	1.869	1.869	1.869	1.869	1.869
X		0.008	0.001	0.001	0.001	0.001	0.001

Table 6 A comparison between the sensors suggested in this study and those found in the existing literature

	RI (n)	FOM(RIU ⁻¹)	(S)	Materials	Application
Ref [65]	1.375–1.550	-	90.10%	Topas	Illegal chemical; sensing
Ref [66]	1.33–1.366	-	91.25%	Zeonex	Ethanol detection
Ref [67]	1.5–1.7	51/RIU	1.036 THz/RIU	Graphene	Biosensing
Ref [68]	1–1.451	27	11.05 THz/RIU	Graphene	Biosensing
Ref [69]	1–1.5	3.18	1.10 THz/RIU	Graphene	Biochemical sensing
Ref [70]	1–1.6	23.61	1.77 THz/RIU	Graphene	Terahertz sensing
Proposed sensor	1–2	0.041	4556 GHz/RIU	SrTiO ₃ , Au graphene, and Bp	Virus sensing

achieved. Changing μ_{c1} to 0.1 eV while keeping μ_{c2} at 0.9 eV (as shown in Fig. 10b clearly reduces transmittance to almost zero. Conversely, setting μ_{c1} to 0.9 eV and μ_{c2} to 0.1 eV

results in a close perfect transmittance again as depicted in Fig. 10a. These results are significant for potential applications in scenarios involving 2-bit encoding.

**Fig. 10** The encoding capability of the proposed sensor

Conclusion

This paper introduces a biosensor based on metasurfaces for virus sensing within the terahertz regime. The sensor incorporates advanced nanomaterials such as graphene, black phosphorus, gold, and SrTiO_3 and has undergone comprehensive design and analysis. Through extensive simulations and optimizations using COMSOL Multiphysics version 6.2, the biosensor shows impressive sensitivity to changes in refractive index (RI) caused by different viruses. It achieves a maximum sensitivity of 4556 GHzRIU^{-1} . Additionally, it demonstrates a figure of merit of 8.499 RIU^{-1} , a quality factor of 1.131, and a minimum detection limit of 0.149. The high sensitivity facilitates quick detection of various viruses based on their unique refractive indices. The assessment of electric field distribution proves quite useful in understanding the sensor's working in conjunction with terahertz waves, with peak absorption established at 0.4 THz. Moreover, the design holds the potential for 2-bit encoding to add to the scheme's function beyond sensing and create more pathways in data encoding and transmitting. When considering the performance characteristics of the sensor, the results are quite satisfactory. However, combining robust nanomaterials with metasurfaces' principles offers a more effective method for rapid, label-free virus detection. As for future work, conducting experiments to validate the proposed sensor design and exploring other promising nanomaterials for even higher sensitivity are possible avenues. Additionally, studying the sensor's efficiency in real-life scenarios is crucial. Miniaturization and integration into portable devices are also areas for further research, as they would enhance the sensor's portability and utility, particularly for point-of-care testing.

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Availability of Data and Materials The data supporting the findings in this work are available from the corresponding author with reasonable request.

Declarations

Ethical Approval Not applicable.

Competing Interests The authors declare no competing interests.

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